Week 7 — Examination Questions (Set 2)

Transformer Architecture, BERT, GPT and Large Language Models Total:

30 marks Suggested duration: 50–60 minutes Level: BSc Level 3

Instructions: Answer all questions. Show reasoning where requested. Marks are indicated in brackets.

Question 1 — Transformer Representations (5 marks)

a) Explain the difference between token embeddings and positional embeddings in a Transformer. (2 marks)

b) A sentence contains the words “dog bites man”. Explain why a Transformer needs positional information to distinguish it from “man bites dog”. (2 marks)

c) State one benefit of processing input tokens in parallel. (1 mark)

Question 2 — Components of an Encoder Layer (5 marks)

A Transformer encoder layer contains multi-head self-attention, a position-wise feed forward network, residual connections and layer normalisation.

a) Explain why multiple attention heads may be more useful than a single attention head. (2 marks)

b) Explain how the position-wise feed-forward network processes the sequence. (2 marks) c)

State one reason for including residual connections. (1 mark)

Question 3 — BERT and Masked Language Modelling (5 marks) During

pretraining, BERT is given: “The bank approved the [MASK] application.”

a) Name BERT’s principal pretraining objective illustrated by this example. (1 mark) b) Explain

how BERT uses the surrounding words to predict the masked token. (2 marks)

c) A bank wants to categorise incoming emails into “loan enquiry”, “card issue” and “complaint”. Explain how BERT could be used for this task. (2 marks)

Question 4 — GPT Generation and Decoding (5 marks)

A GPT model is prompted with “The meeting starts at”. Its next-token probabilities are:

Candidate token	Probability

nine 0.45
ten 0.35

noon 0.15
tomorrow 0.05

a) Which token will greedy decoding select? (1 mark)

b) Explain why sampling might produce a different token from greedy decoding. (2 marks)

c) Describe the two steps GPT repeats to produce a longer response after selecting its first token. (2 marks)

Question 5 — Architecture Selection: Applied Scenario (6 marks)

A university wishes to implement three applications: (A) identify student names and dates in administrative emails; (B) translate announcements from English to French; (C) draft personalised replies to student enquiries.

For each application, recommend an appropriate Transformer architecture (encoder-only, encoder–decoder, or decoder-only) and justify the choice. (2 marks per application; 6 marks total)

Question 6 — Base GPT, Instruction Tuning and Inference (4 marks)

a) Explain one difference between the objective of base GPT pretraining and instruction tuning. (2 marks)

b) A deployed instruction-tuned LLM answers a user question. State whether its model parameters are normally updated during this inference and explain why. (2 marks)

TOTAL: 30 MARKS
SIS 3204Y(5) | Natural Language Processing | Week 7

Set 2 — Model Answers and Marking Rubric

Award credit for technically correct equivalent answers. Do not require exact wording. Accept reasonable alternatives when justified.

Question 1 — Transformer Representations (5 marks)

Part	Expected answer / mark allocation
a	Token embeddings encode learned token identity/semantic features (1); positional embeddings/encodings represent token position or order (1).
b	The sentences contain the same words (1) but different orders and meanings; positional information distinguishes

their arrangements (1).
c Parallel processing can improve training throughput/efficiency compared with strictly sequential token processing.
Marks 2

1

2

Question 2 — Components of an Encoder Layer (5 marks)

Part	Expected answer / mark allocation	
		two linear transformations with a non-linear activation
a	Different heads can learn different relationships or attend to different positions (1); combining heads gives richer contextual representations (1).	
b	An FFN applies the same learned transformation to	between them (1). **Marks** 2
	each token position independently (1), typically	2

c Residual connections 1	preserve information and/or improve gradient flow, helping train deep networks.

Question 3 — BERT and Masked Language Modelling (5 marks)

Part	Expected answer / mark allocation
a Masked language modelling (MLM).	
b BERT uses bidirectional self-attention, accessing tokens before and after [MASK] (1), and uses the combined context to predict a plausible token such as “loan” (1).	
c Use a pretrained BERT encoder and attach a classification head, often on [CLS] (1); fine-tune on labelled emails to predict the three categories (1).	

Question 4 — GPT Generation and Decoding (5 marks)

Part	Expected answer / mark allocation
a nine (highest probability, 0.45).	
b Greedy decoding always picks the most probable token (1); sampling draws from the probability distribution and can choose another candidate, such as “ten” (1).	
c Append the selected token to the context (1); run next token prediction again and repeat until a stop token or other stopping criterion (1).	

Marks 1

Question 5 — Architecture Selection (6 marks)

Part Expected answer / mark allocation

2

2

Marks 1

2

2

Marks 2

A Encoder-only, e.g. BERT (1); bidirectional token representations suit entity extraction/NER (1).

	Encoder–decoder, e.g. T5/Marian (1); encoder reads the source and decoder generates the translated target with cross-attention (1).	2
C Decoder-only, e.g. instruction-tuned GPT (1); autoregressively generates a contextual reply one token at a time (1).		2

Question 6 — Base GPT, Instruction Tuning and Inference (4 marks)

Part	Expected answer / mark allocation	
a	Base GPT pretraining learns next-token prediction on general text (1); instruction tuning further trains the model on instruction–response examples to better follow requests (1).	2
b	Parameters are normally fixed during inference (1); the model computes an answer from its existing learned weights without a training/optimisation update (1).	2
Marks		4

Marking note: In Question 5, alternative architectures can be accepted where the student gives a technically defensible explanation (for example, an instruction-tuned decoder-only model for

translation).

Assessment Blueprint

Content area	Question	Marks
Embeddings, positions and parallelism	Q1	5
Encoder components	Q2	5
BERT and MLM	Q3	5
GPT decoding	Q4	5
Architecture selection	Q5	6
Instruction tuning and inference	Q6	4
TOTAL		30